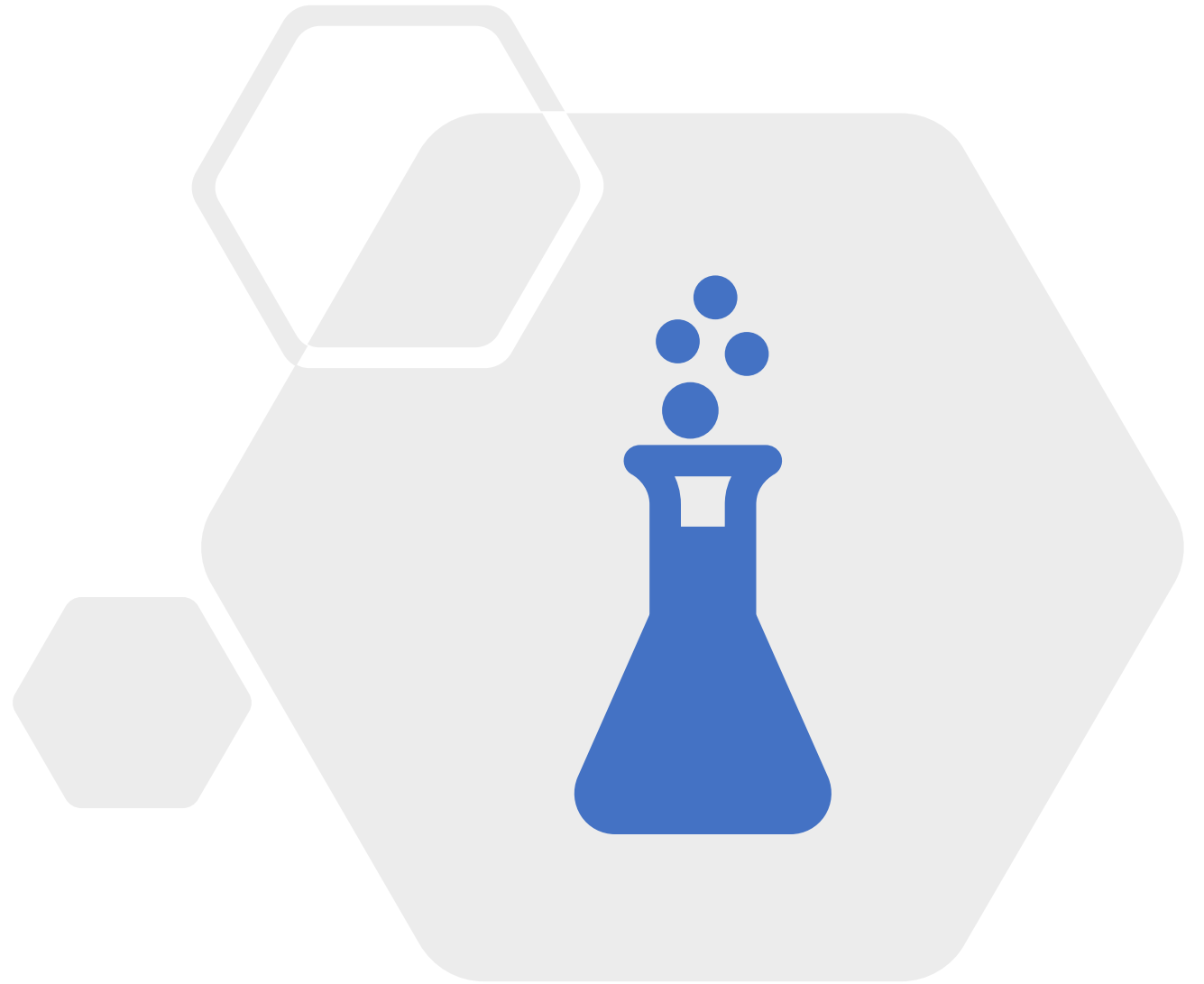
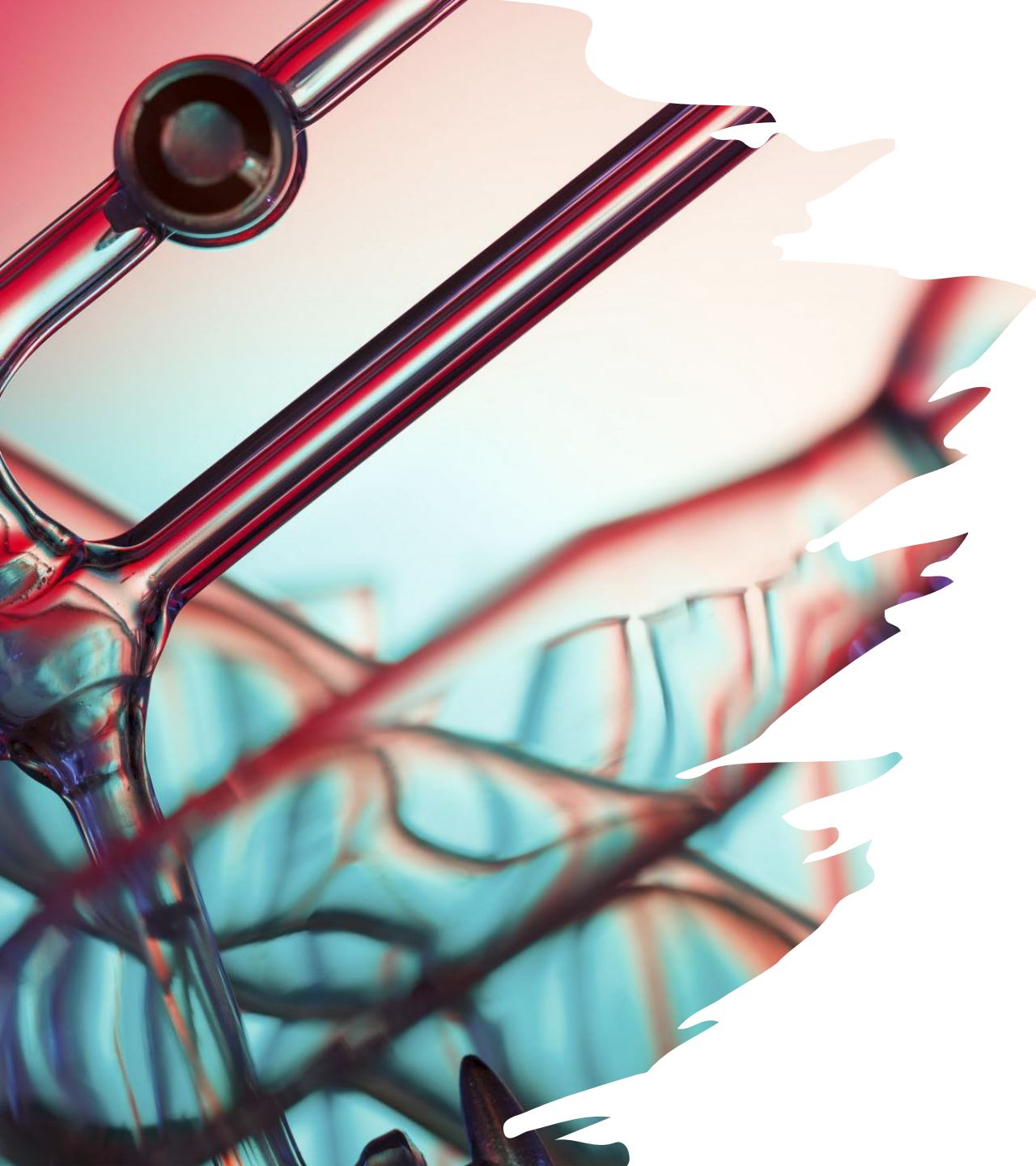


Experimental Research



Roadmap: Three qualities of Experimental Studies

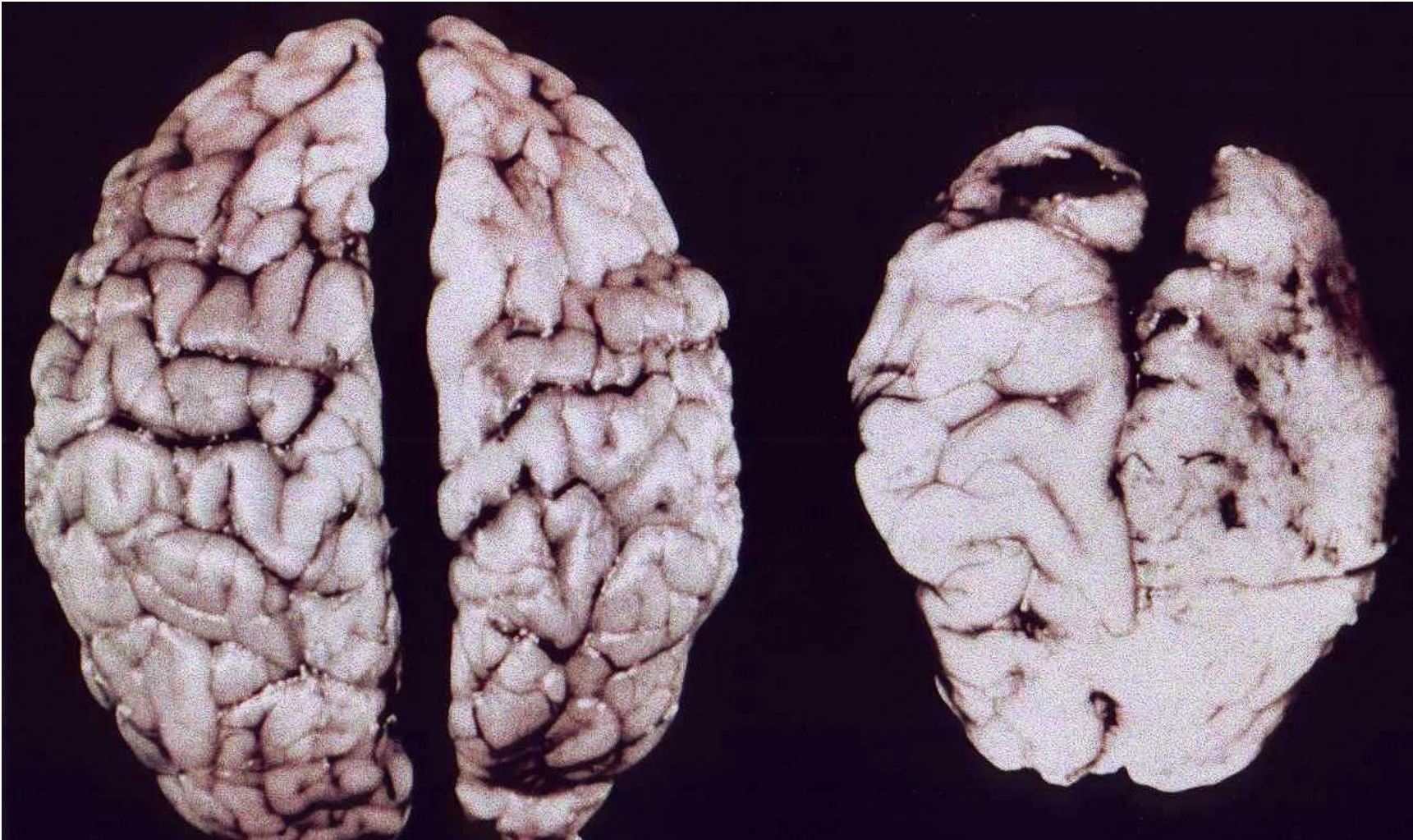
1. Researcher manipulates one variable.
2. Researcher must control the group assignment
3. Researcher must control extraneous variables that influence Ss' behavior.



Manipulating One Variable

In experimental studies, you are interested in how one (or more) variable changes behavior.

(not just how they are related).



6-week-old baby without
Fetal Alcohol Syndrome

6-week-old baby with
Fetal Alcohol Syndrome

H_1 : Alcohol causes brain abnormalities in the developing fetus.

Vary or Manipulate amount of Alcohol

Groups get either:

0.6g/day, 0.3g/day, 0g/day.



Each variation in the IV is a “level”.

Independent Variable -- The researcher is setting or controlling the variation.

The amount of alcohol is independent of everything else in the study.

Dependent Variable – Response being measured

Brain size, white matter to gray matter ratio, density, gyri, ventricle size



0.6g/day, 0.2g/day, 0g/day

Groups that receive some amount of alcohol:

Experimental Groups

Group that does not receive any alcohol:

Control Group

Why a control group?

Baseline

Levels can vary in two ways:

Quantitative Differences

– IV varies in quantity.

Ex: *grams/kg of Alcohol*

Qualitative Differences

- IV varies in type.

Ex: Environment
Instructions
Diet

Types of IVs

Environmental Manipulations

modifications of the participant's physical or social environment, may involve **confederates**

Instructional Manipulations

vary the independent variable through the verbal instructions that participants receive

Invasive Manipulations

create physical changes in the participant's body through surgery or the administration of drugs

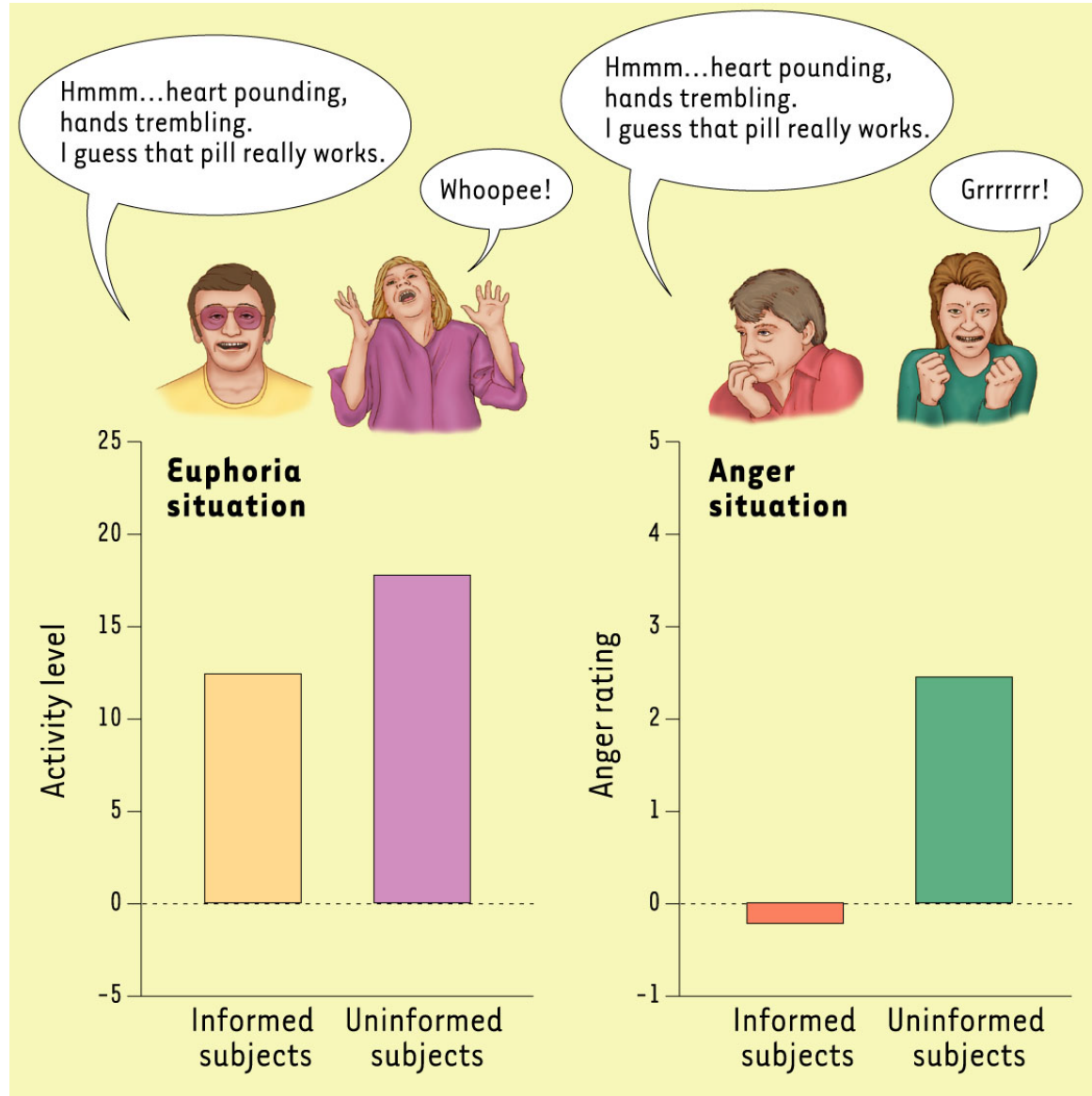
Schachter & Singer

Subjects left in room with confederates that were either: Euphoric (happy & silly) acting OR Angry (offended by questions)

Told the drug would make aroused OR given no information

Who would be influenced by the confederate's mood?

Patients injected with a stimulant (adrenaline) OR saline.



How good is your IV?

A good IV has enough levels to detect changes in behavior.

To determine:

Pilot Test (pre-study)

- measure levels of IV for response

Manipulation Check (post-study)

- ensure that the level of your IV was effective



A pharmaceutical company developed a new drug to relieve depression and hired a research organization to investigate the potential effectiveness of the drug. The researchers contacted a group of psychiatric patients who were experiencing chronic depression and randomly assigned half of the patients to the drug group and half of the patients to the placebo group.

What is the independent variable?

How many levels does it have?

Was there a control group?

What is the dependent variable?

An investigator hypothesized that people in a fearful situation desire to be with other individuals. To test this hypothesis, the experimenter randomly assigned 50 participants to either a high or low fear group. Participants in the low fear group were told that they would be shocked but that they would experience only a small tingle that would not hurt. Participants in the high fear group were told that the shock would be quite painful and might burn the skin without permanent damage.

What is the independent variable?

How many levels does it have?

Was there a control group?

What is the dependent variable?

A study was conducted to investigate the hypothesis that watching televised violence increases aggression in children. Fifty kindergarten children were randomly assigned to watch either a violent or a nonviolent television program. After watching the televised program, the children were allowed to engage in an hour of free play while trained observers watched for aggressive behavior and recorded the frequency with which aggressive acts took place.

What is the independent variable?

How many levels does it have?

Was there a control group?

What is the dependent variable?

Three qualities of experimental studies

Researcher manipulates one variable.

Researcher must control the group assignment

Researcher must control extraneous variables that influence Ss' behavior.



Wrapping Up...



This Time...
Control the group
assignment

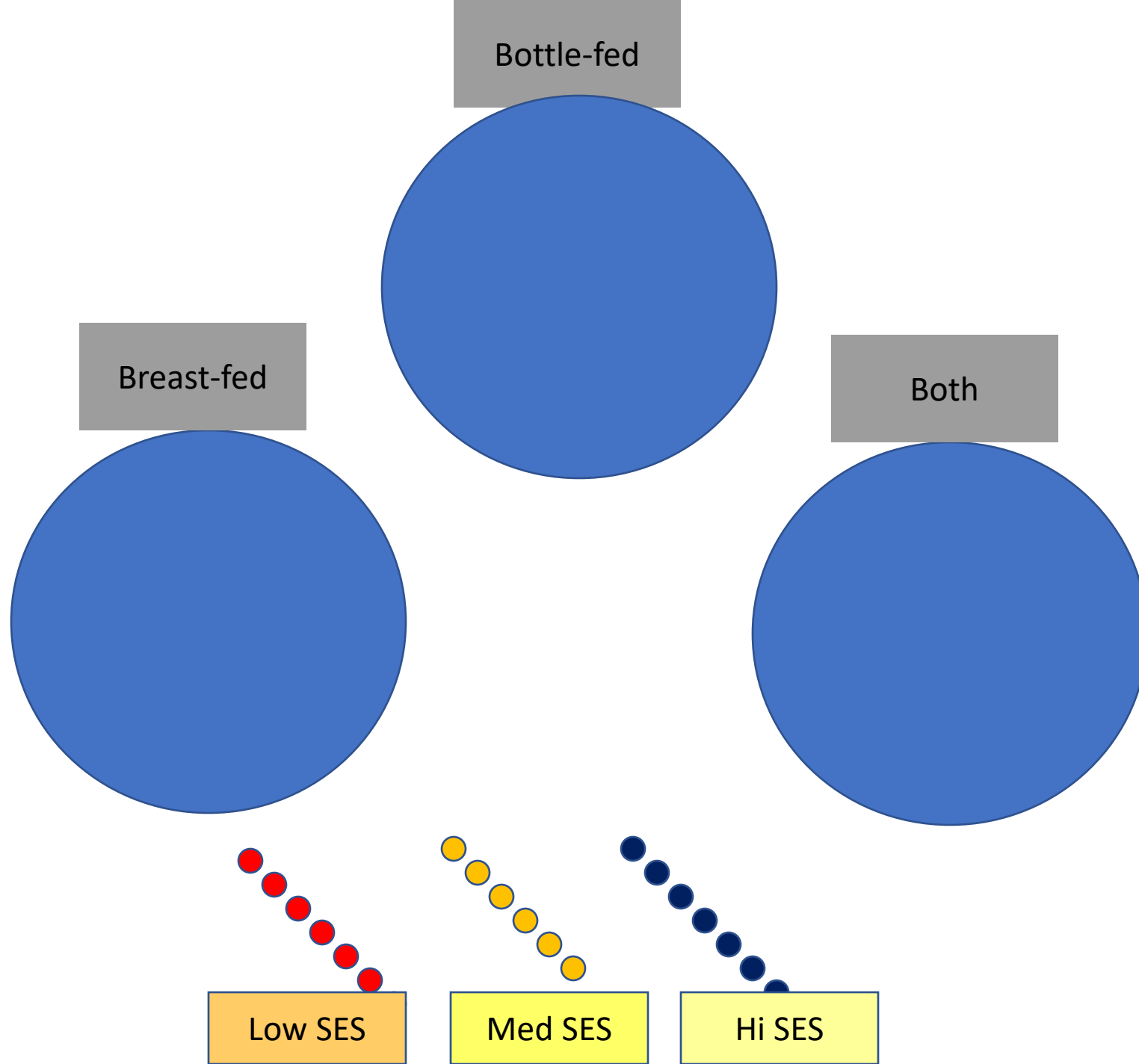


2. Control the group assignment

Ss must be assigned to groups, so they are **initially equivalent**.

If groups differ in anyway *before* the IV is given, then cannot tell if variation in behavior is caused by:

- the IV, OR
 - pre-existing differences.
- 



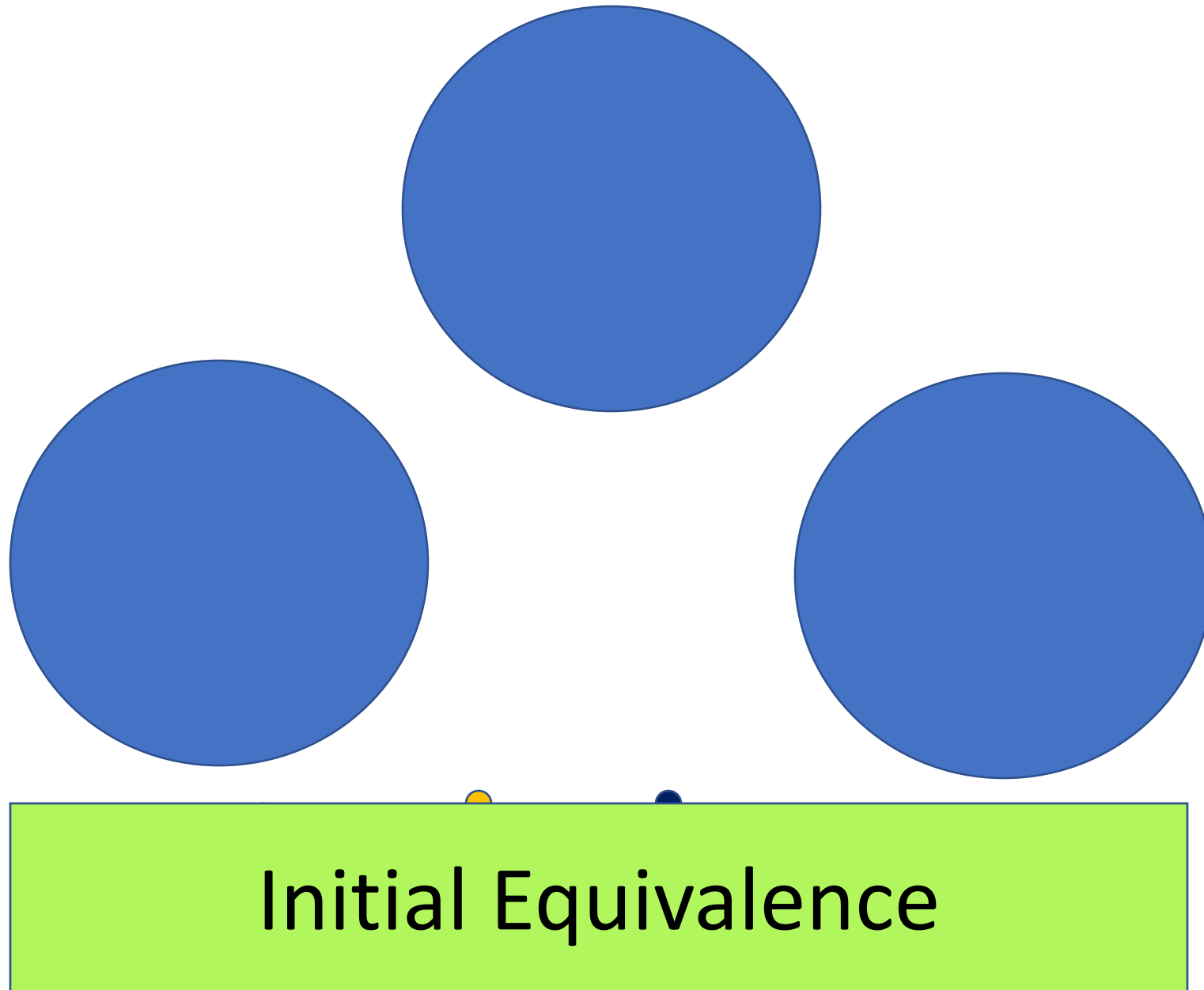
We need to
RANDOMLY assign
Ss to levels of the IV.

This ensures that ON AVERAGE, Ss
in your groups will not differ.

There is any equally likely chance of
getting into each level of your IV.



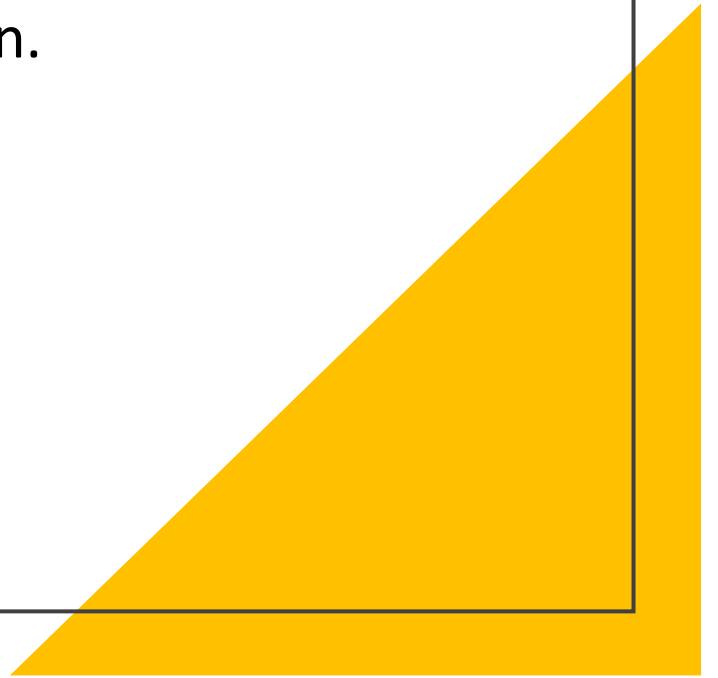
FIGURE 10.3



Simple Random Assignment

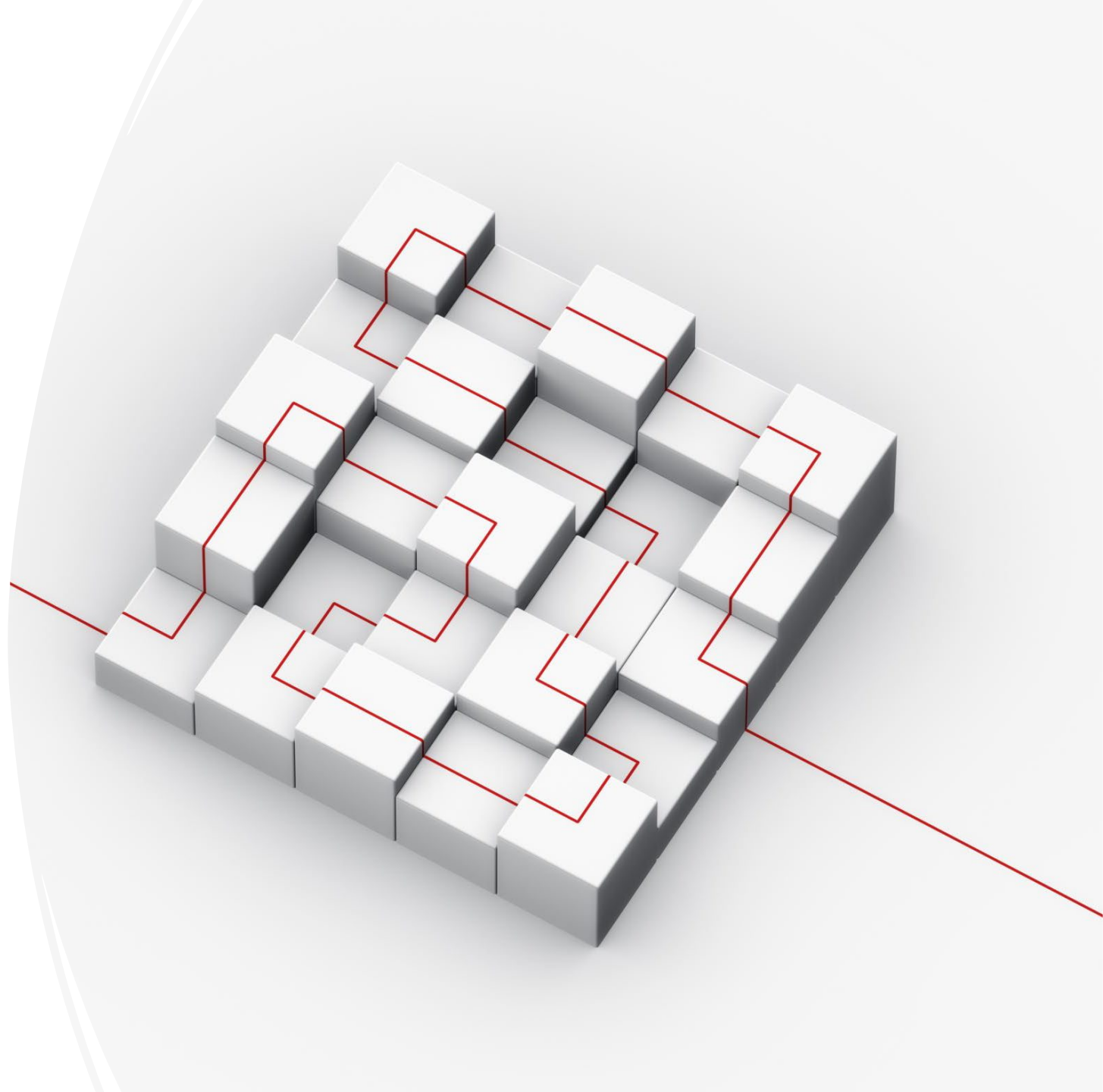
Placing Ss in conditions so that every subject has an equal probability of being placed in any experimental condition.

Example: Coin Flip



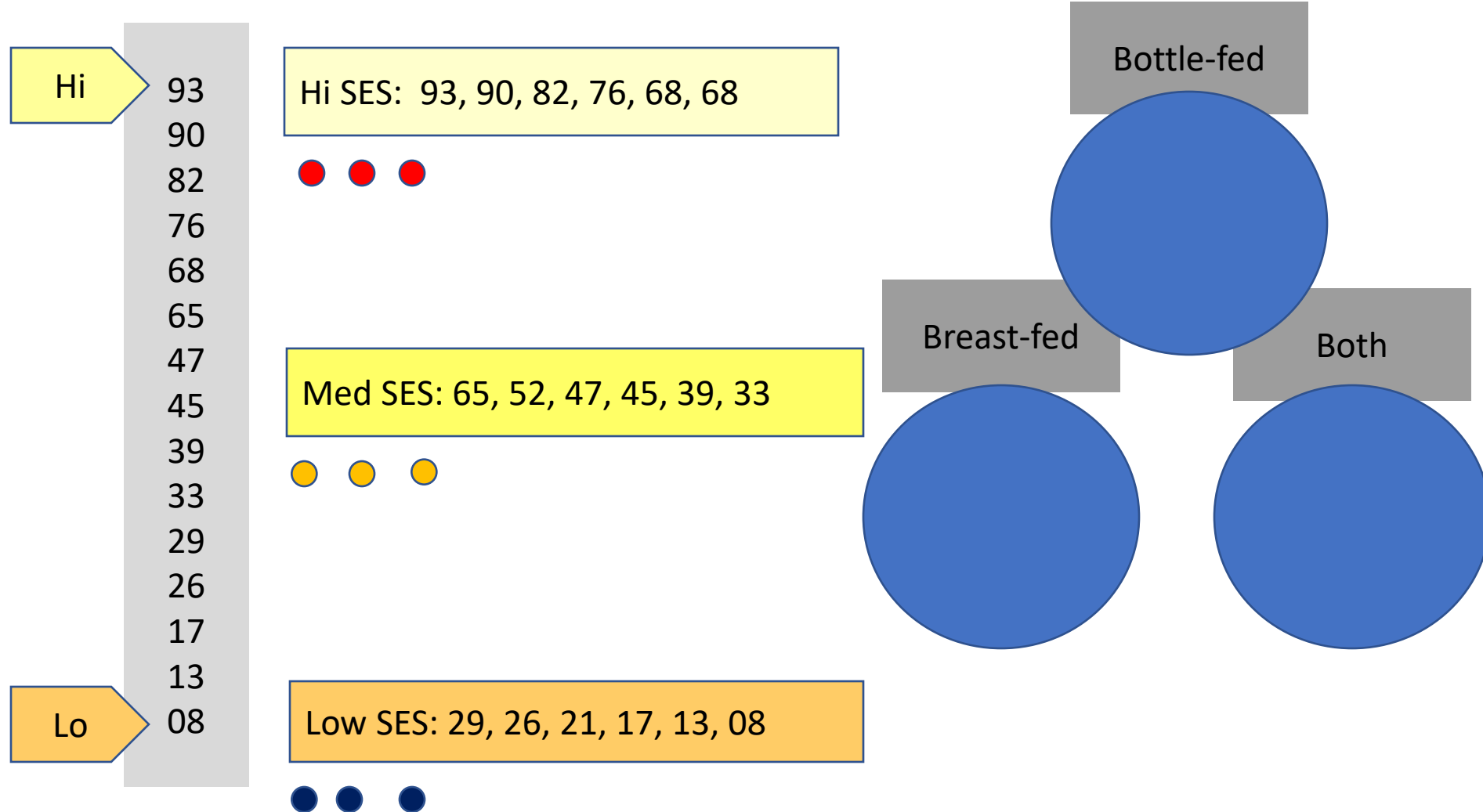
Matched Random Assignment

- Helps to ensure that the conditions will be similar along some specific dimension,
 - such as age or intelligence.
- Participants are matched
 - into homogenous blocks by order on a measure, and then
 - participants within each block are assigned randomly to conditions



n=18 mother & infant pairs each with SES index

RAW data: 17, 82, 65, 33, 68, 29, 93, 13, 47, 68, 21, 90, 45, 39, 26, 08, 52, 76



3 IVs

Environmental Manipulations

modifications of the participant's physical or social environment, may involve **confederates**

Instructional Manipulations

vary the independent variable through the verbal instructions that participants receive

Invasive Manipulations

create physical changes in the participant's body through surgery or the administration of drugs

Repeated Measures Design

Each subject gets to be in each condition.

- Within-Subjects Design

Do you need random assignment?

Do you have Initial Equivalence?

No, this design eliminates the need for random assignment because every subject is the same in each groups.

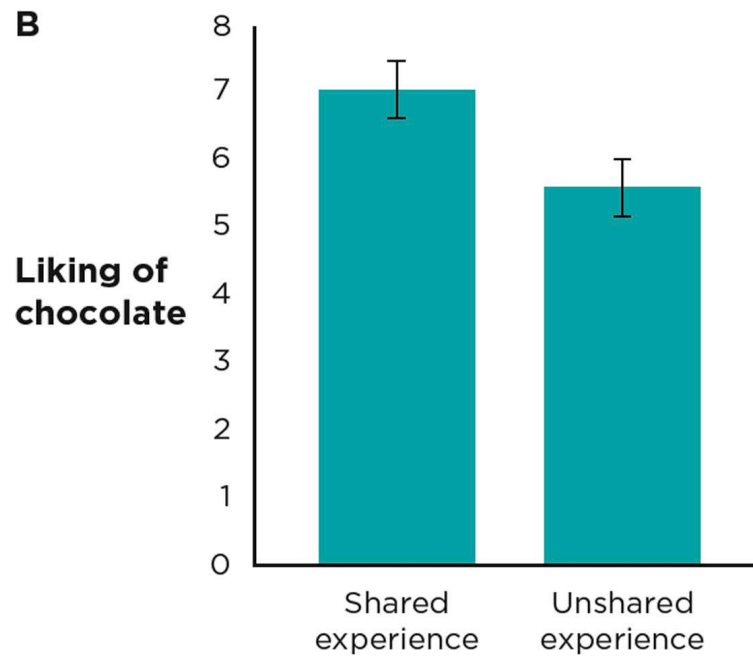
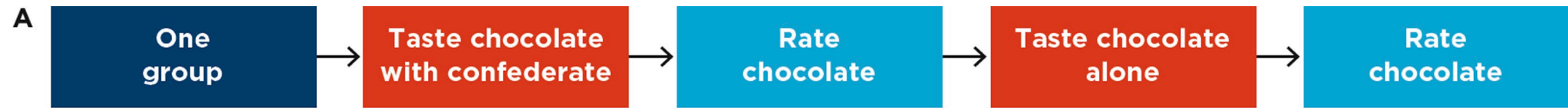


FIGURE 10.14

Power

The ability to detect differences between the levels of the IV,

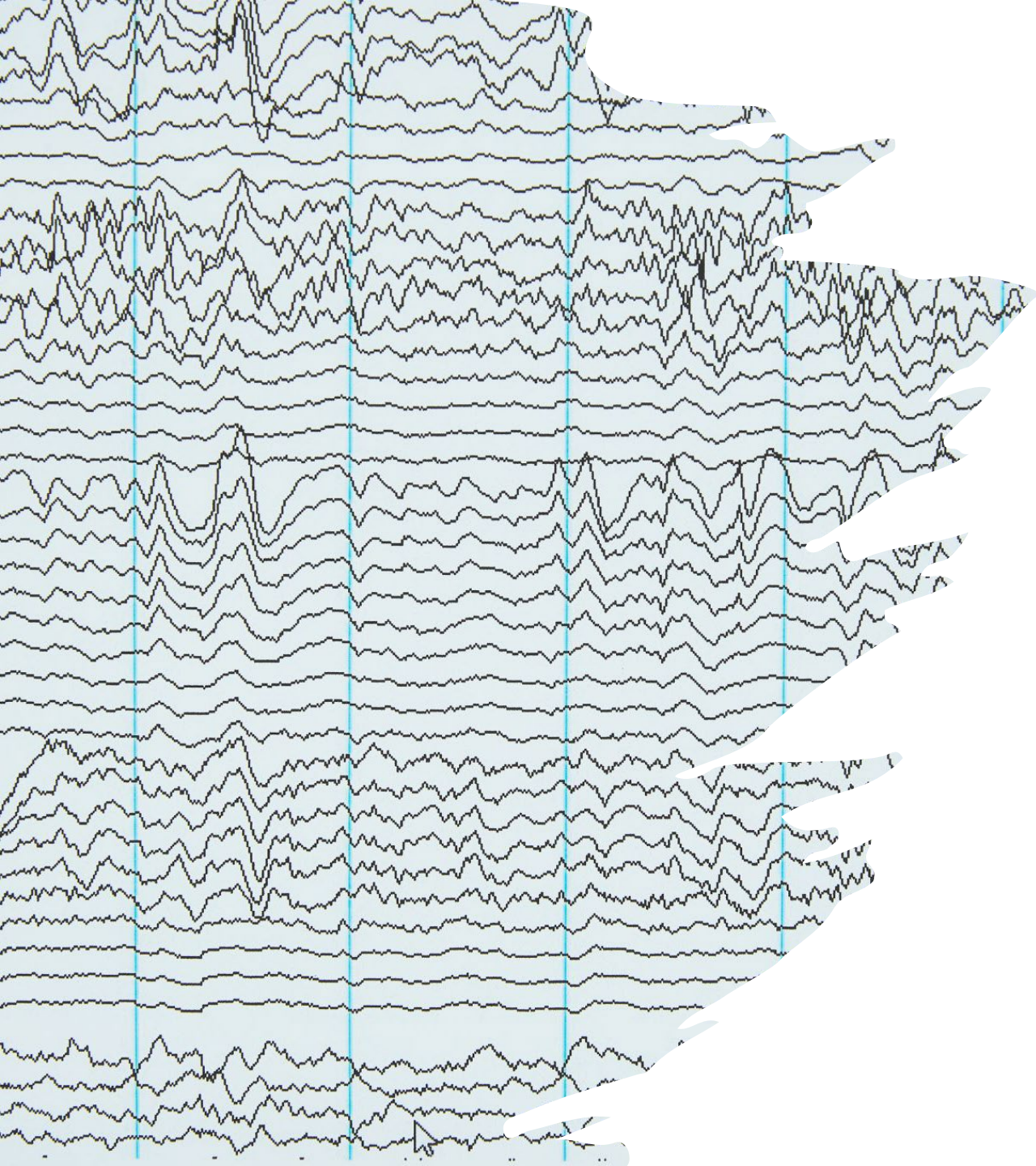
- IF there are differences to detect.

Within Ss > POWER > Between Ss

If your alternative hypothesis is correct, more likely to detect the truth.

Why not always use Repeated Measures?

- Order effects
 - The order in which levels of the IV are received affects behavior.
- Practice? Fatigue
 - Ss performance improves because they have completed the dependent variable several times/
 - Ss become tired, bored, or less motivated as the experiment progresses.
- Carryover
 - Effects on one condition carry over to the next condition
- Solution?
 - Full counterbalancing & Latin Square



Full Counterbalancing

- Presenting levels of the IV in different orders to different participants,
 - when possible, all orders are used.
- Core Idea:
 - Every possible order exists and every level of the IV appears in each position an equal number of times.

Full Counterbalancing Example

Measure Consumer purchases when offering different reward programs:

- 1.) Immediate reward IR
- 2.) Short-term reward STR
- 3.) Long-term reward LTR
- 4.) No rewards Control

Every possible order exists and every level of the IV appears in each position an equal number of times.

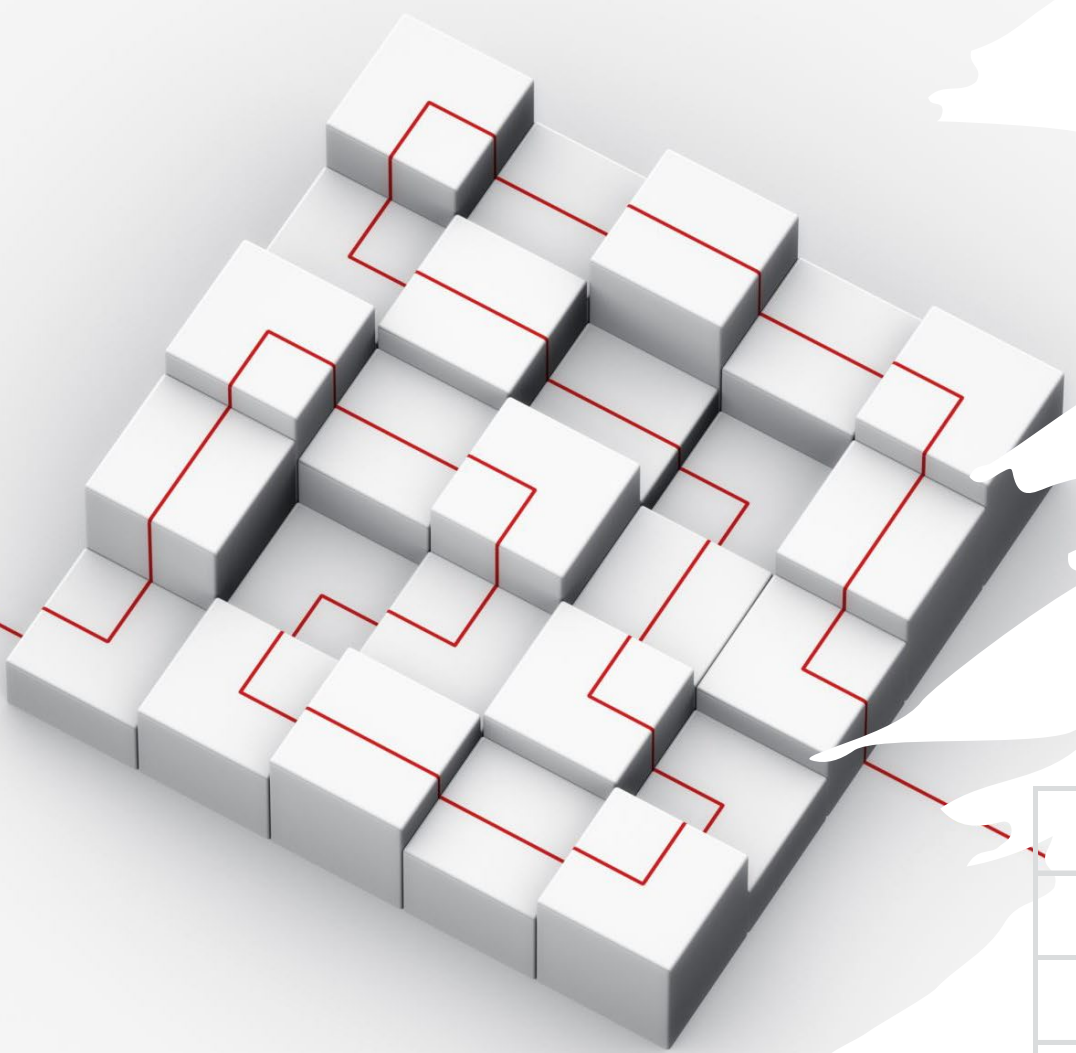


	1st Order	2nd Order	3rd Order	4th Order
1	Control	IR	STR	LTR
2	Control	IR	LTR	STR
3	Control	STR	IR	LTR
4	Control	STR	LTR	IR
5	Control	LTR	IR	STR
6	Control	LTR	STR	IR
7	IR	Control	STR	LTR
8	IR	Control	LTR	STR
9	IR	STR	Control	LTR
10	IR	STR	LTR	Control
11	IR	LTR	Control	STR
12	IR	LTR	STR	Control
13	STR	Control	IR	LTR
14	STR	Control	LTR	IR
15	STR	IR	Control	LTR
16	STR	IR	LTR	Control
17	STR	LTR	Control	IR
18	STR	LTR	IR	Control
19	LTR	Control	IR	STR
20	LTR	Control	STR	IR
21	LTR	IR	Control	STR
22	LTR	IR	STR	Control
23	LTR	STR	Control	IR
24	LTR	STR	IR	Control

Latin Square Design

Each condition appears once at each ordinal position, each precedes and follows every condition once.

Participants are randomly assigned to four groups, each with a different order of conditions.



	1st Order	2nd Order	3rd Order	4th Order
Group 1	Control	Immed	Long	Short
Group 2	Immed	Short	Control	Long
Group 3	Short	Long	Immed	Control
Group 4	Long	Control	Short	Immed



Wrapping Up...



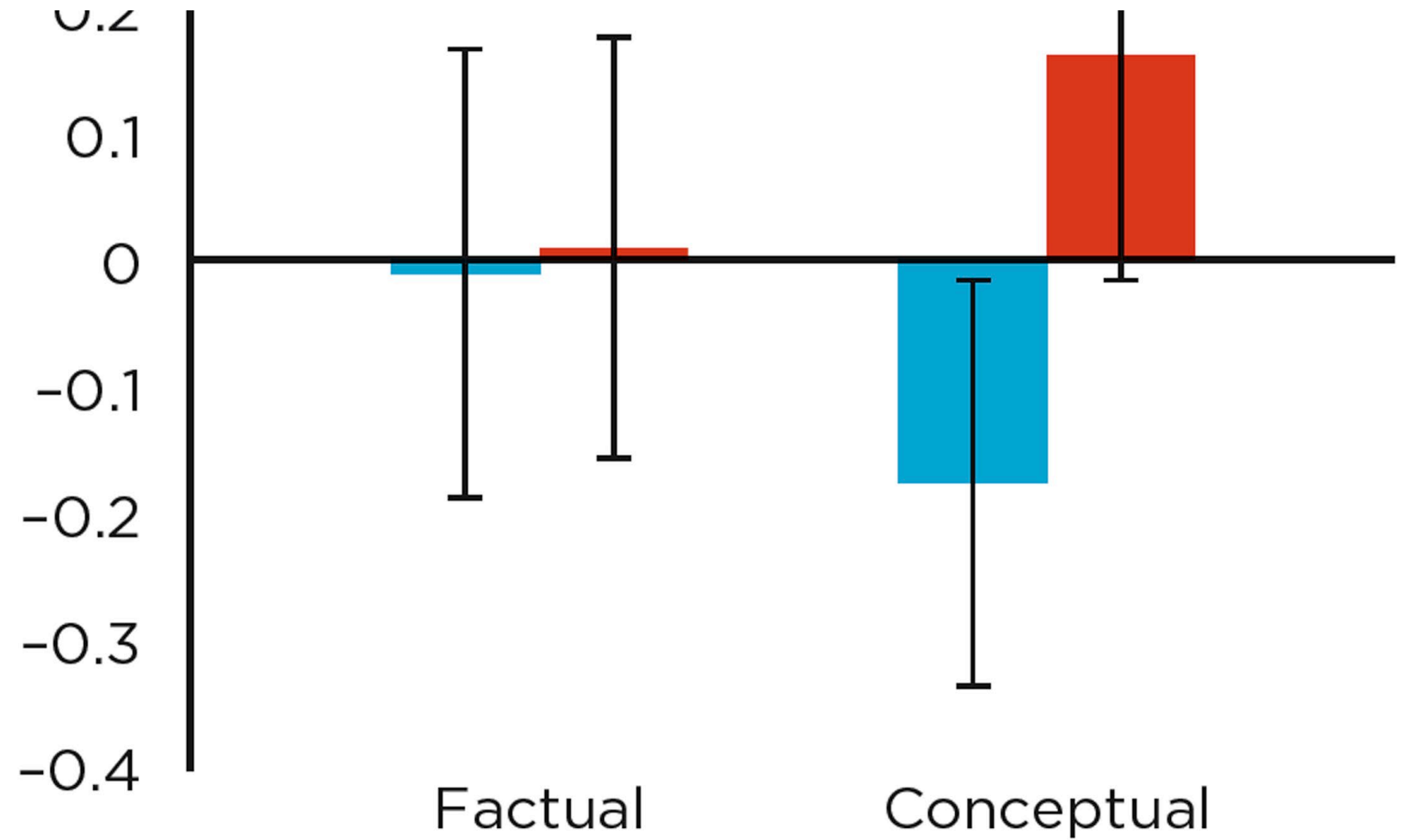


This Time...
Control
Extraneous
Variables

Three qualities of experimental studies

1. Researcher manipulates one variable.
 2. Researcher must control the group assignment
 3. Researcher must control extraneous variables that influence Ss' behavior
- Control *what*?
 - Everything that is not of interest in your study.

**Score on
essay
questions
(standardized)**

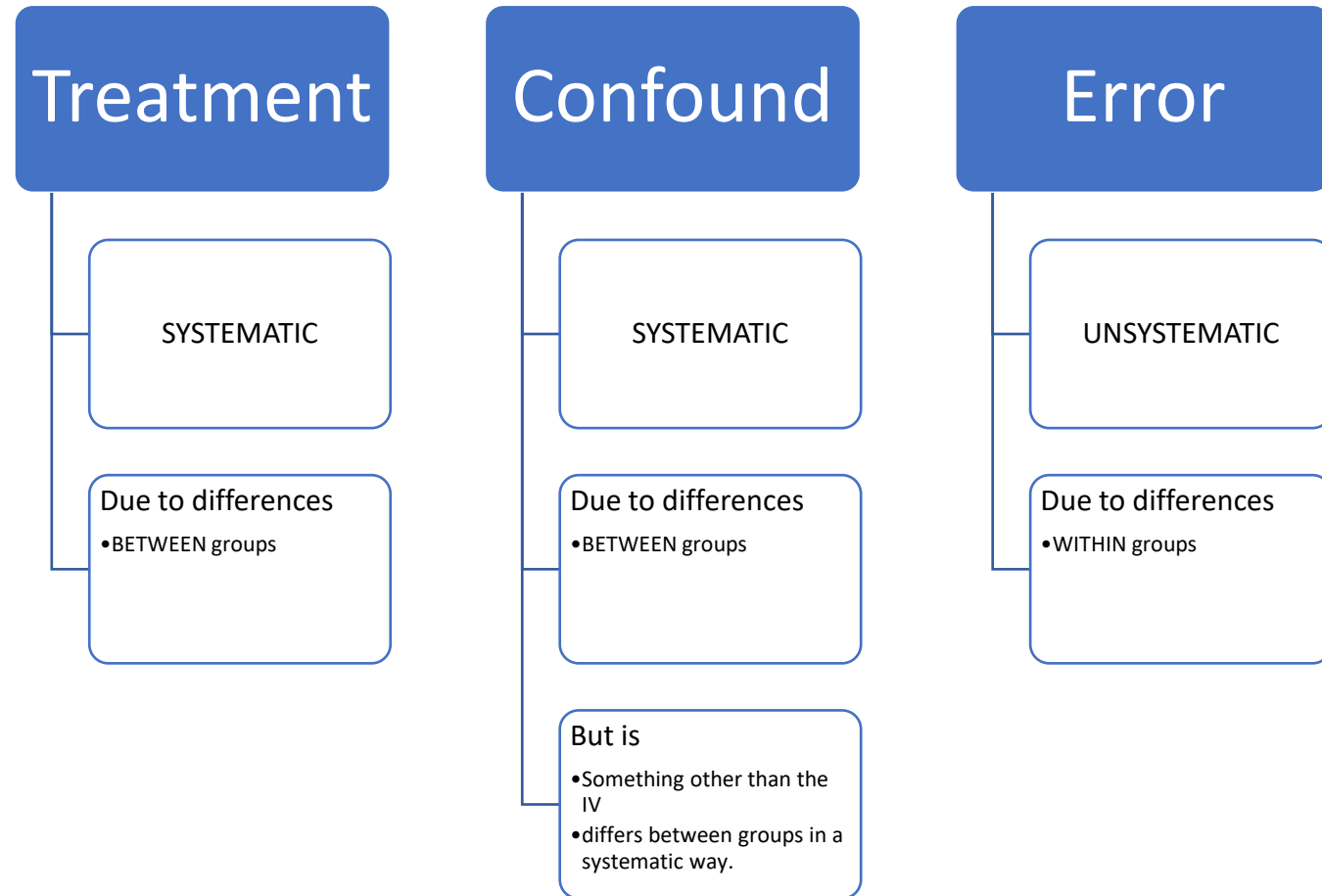




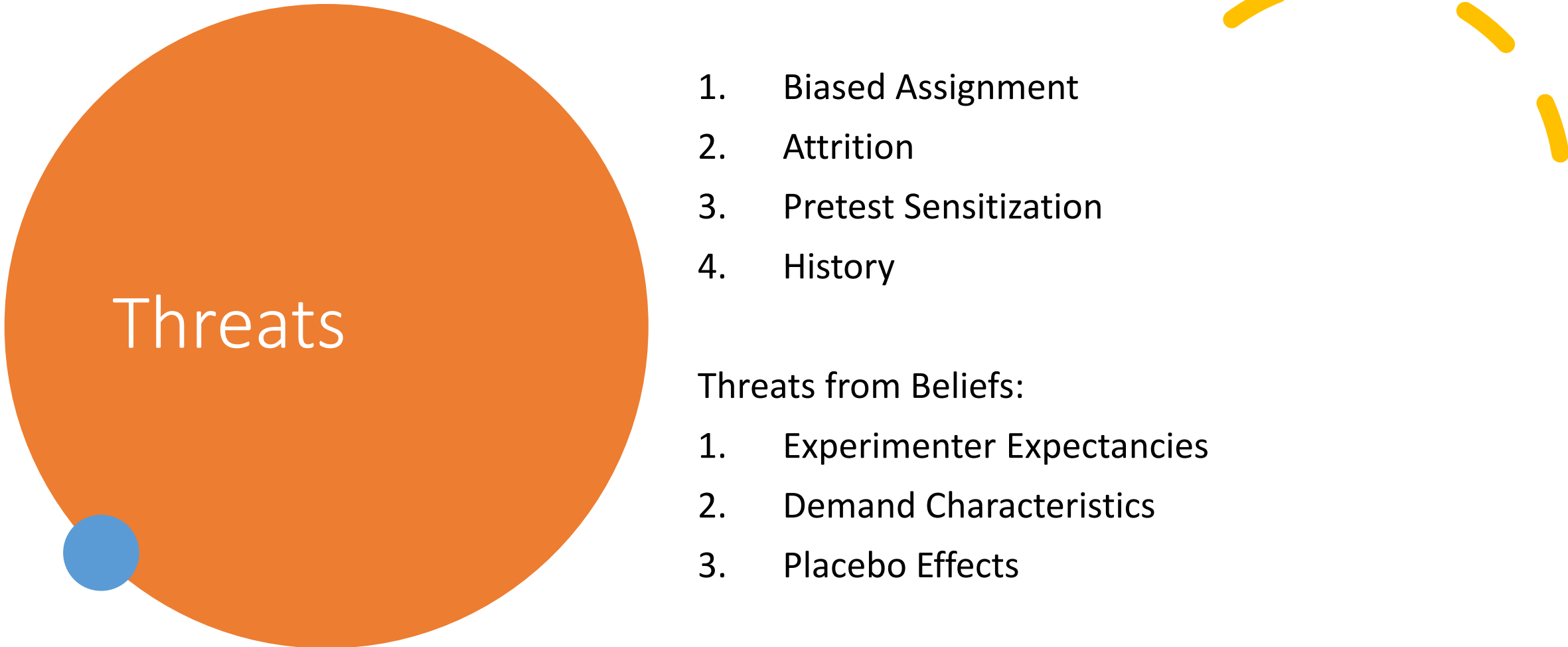
Is any of the
total variability
in observed in
scores due to
the IV?

- Total Variance = treatment + confound + error
- Confounds must be addressed
 - because an experimental researcher cannot tell
 - if differences between groups are due to
 - IV or the
 - confound.
 - Eliminated thru experimenter control.

Total Variance ...



- 
- A problem with confounds is a problem with
 - **Internal Validity**
 - The degree to which a researcher draws accurate conclusions about the effects of the IV.
 - Know and Eliminate your Enemy
 - (in Research Design)...
- 

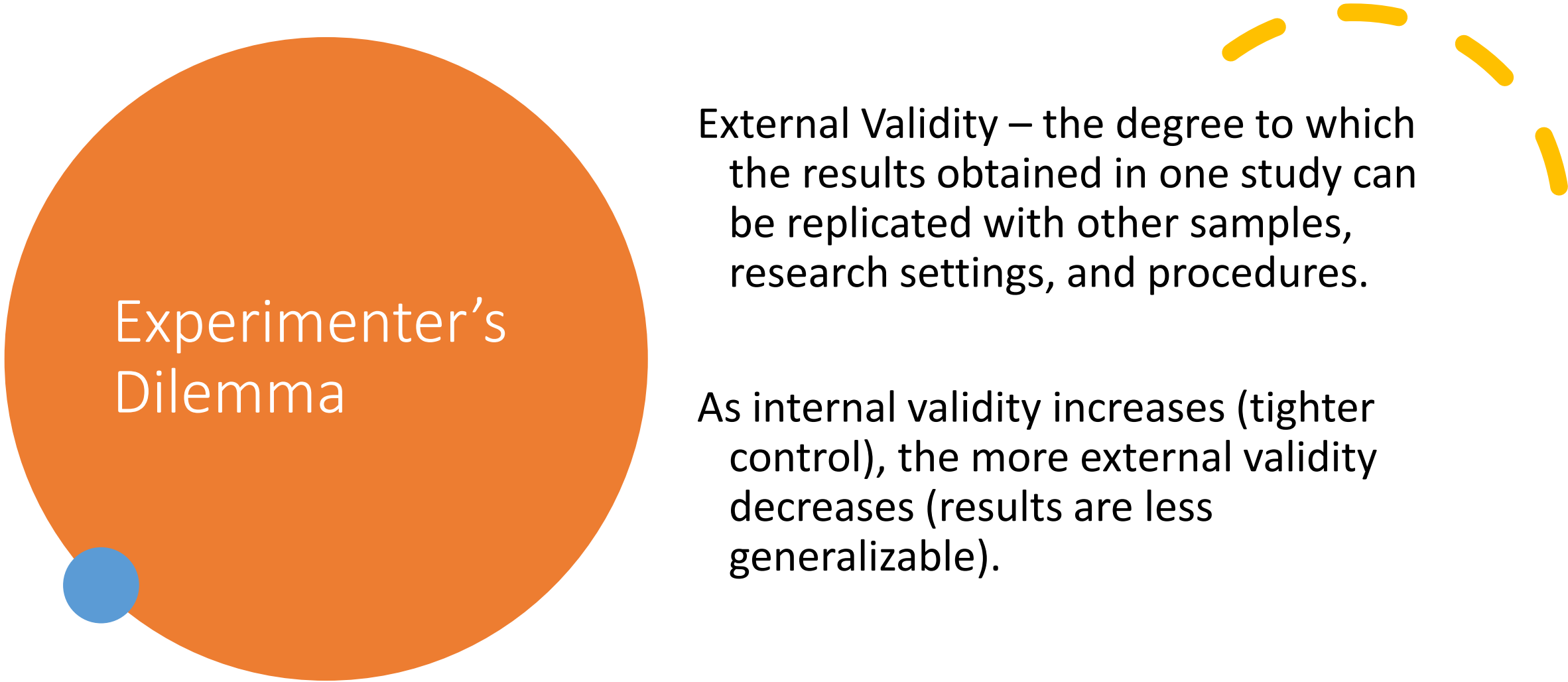


Threats

1. Biased Assignment
2. Attrition
3. Pretest Sensitization
4. History

Threats from Beliefs:

1. Experimenter Expectancies
2. Demand Characteristics
3. Placebo Effects



Experimenter's Dilemma

External Validity – the degree to which the results obtained in one study can be replicated with other samples, research settings, and procedures.

As internal validity increases (tighter control), the more external validity decreases (results are less generalizable).



Manipulate, Control, Control....

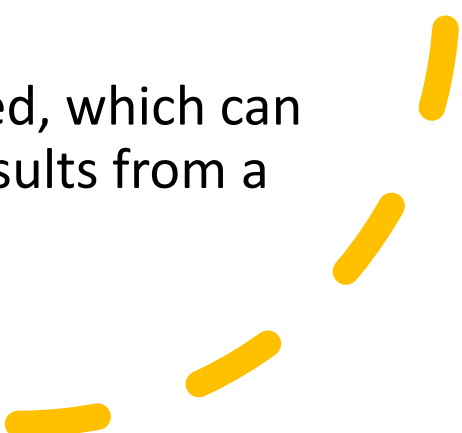
The more tightly controlled, the more artificial.

“But, that study is nothing like the real world...”

NOT A VALID CRITICISM

Experiments should *never* be generalized. Remember, we are testing hypotheses in order to support theories. Studies are done to *isolate* the factors causing behavior, thought, experience.

REPLICATION allows theories to be supported, which can then generalize the theory, but not the results from a single study.





Wrapping Up...